

CS331 Topology Summary Sheet

Your Chapter 1 slides define **topology** as how a network is physically built or how devices are interconnected. The four basic topologies in your course are **Mesh, Star, Bus, and Ring**, and they can be combined into a **Hybrid topology**.

Topology	Structure	# Links	Advantages	Disadvantages
Mesh	Every device has a dedicated point-to-point link to every other device	$\frac{N(N-1)}{2}$	Very robust ; very secure ; no traffic/congestion problems on dedicated links; easy to locate faulty links	Very expensive ; lots of wiring; difficult to install/update; each device needs N-1 ports
Star	Every device connects directly to one central device	N	Easy to install; easy to add/remove devices; cheaper than mesh; robust against individual link failure; easy fault detection; relatively secure	Central device = single point of failure ; if it fails, entire network stops
Bus	All devices connect to one main backbone cable using drop lines	1 main + N drop lines	Easy to install; requires less cabling than mesh/star	Difficult fault isolation; difficult to add devices; break in main bus stops all transmission
Ring	Each device connects to exactly two neighboring devices , forming a closed loop	N	Easy to install; less cable; easy to add/remove devices; easy fault isolation	Not robust ; a link/device failure can disrupt the ring
Hybrid	Combination of 2+ topologies	Depends	Can combine strengths of different topologies; flexible for larger/complex networks	Design, installation, troubleshooting, and cost depend on the topologies combined

The mesh details—including the $N(N-1)/2$ formula, robustness, dedicated links, security, and high wiring/port requirements—are directly from your slides. The star topology uses dedicated links to a central controller, with **N links**, but the central controller is its critical failure point.

How to mix topologies — Hybrid Topology

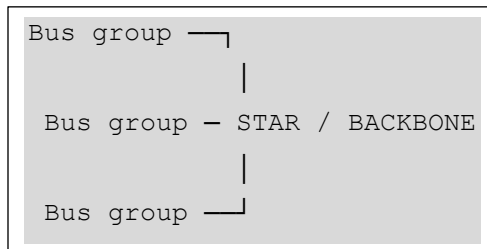
This is the part I would memorize especially well:

Hybrid = combine two or more basic topologies.

Your slides specifically show a **star backbone with three bus networks** and identify **Star–Ring** and **Star–Bus** as hybrid combinations.

Star–Bus

Think:



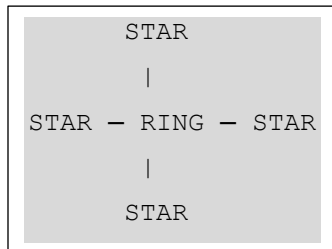
You combine **Star + Bus**.

Why use it? You get the organizational/scalability advantages of star while allowing groups or segments to use a bus structure.

Main weakness: You inherit weaknesses from both. A failure in an individual bus backbone can affect that bus segment, while failure of a critical central device/backbone can affect a much larger portion of the network.

Star–Ring

Think:



Individual groups can use **star topology**, while those groups are interconnected in a **ring**.

Why use it? Star makes individual groups easier to manage, while the ring provides the interconnection between the groups.

Main weakness: You still inherit the relevant failure points of both structures.

General Hybrid Rule

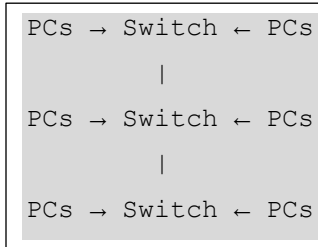
When they give you a diagram, **don't judge the topology only by how the computers are connected locally**.

Look at it in two levels:

Inside each group → identify the local topology.

Between groups → identify the backbone topology.

So:



Each group may be a **star**, while the way those stars are interconnected determines the overall hybrid.

Fast Exam Recognition

If you see...	Think...
Everyone connected to everyone	Mesh
Everyone connected to one center	Star
Everyone attached along one main cable	Bus
Devices connected in a closed loop	Ring
Different topology patterns combined	Hybrid

Failure trick

This is probably the easiest way to distinguish them:

Mesh: one link dies → **network still works**

Star: one device's link dies → **only that device affected**; center dies → **everyone affected**

Bus: main cable dies → **everyone affected**

Ring: ring connection breaks → **network can be disrupted**

Hybrid: ask **which component failed** and **which topology that component belongs to**.

One extra CS331 detail worth remembering: your later connecting-devices slides explicitly note that a **hub can produce a physical star topology while being logically bus**, so **physical topology and logical operation are not always the same thing**.